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import pandas as pd
import matplotlib.pyplot as plt
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```
df = pd.read_csv(r'C:\Users\Muhammad Taha\Desktop\Coffee Shop Sales Analysis.csv')
```

```
print("First look at our data:")
print(df.head())
print("\n")
```

First look at our data:

	Product	Category	Cost	Price	March_Sales	April_Sales	\
0	Latte	Hot Coffee	1.2	4.50	3200	3680	
1	Cappuccino	Hot Coffee	1.1	4.25	2200	2400	
2	Americano	Hot Coffee	0.8	3.50	1800	1900	
3	Drip Coffee	Hot Coffee	0.5	2.75	1200	1100	
4	Espresso	Hot Coffee	0.6	3.00	900	950	

	March_Units	April_Units
0	711	818
1	518	565
2	514	543
3	436	400
4	300	317

```
total_march = df['March_Sales'].sum()
total_april = df['April_Sales'].sum()
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total_growth = ((total_april - total_march) / total_march) * 100
```

```
print(f"Total March Sales: ${total_march:,.0f}")
print(f"Total April Sales: ${total_april:,.0f}")
print(f"Overall Growth: {total_growth:.1f}%")
print("\n")
```

```
Total March Sales: $25,050
Total April Sales: $29,210
Overall Growth: 16.6%
```

```
top_products = df.sort_values('April_Sales', ascending=False).head(5)
print("TOP 5 PRODUCTS BY APRIL SALES:")
print("=" * 40)
for index, row in top_products.iterrows():
    print(f"{row['Product']}: ${row['April_Sales']:,.0f} ({row['Revenue_Share']:,.1f}% of total)")
print("\n")
```

```
TOP 5 PRODUCTS BY APRIL SALES:
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Latte: $3,680 (12.6% of total)
Cappuccino: $2,400 (8.2% of total)
Breakfast Sandwich: $2,350 (8.0% of total)
Cold Brew: $2,200 (7.5% of total)
Iced Latte: $1,950 (6.7% of total)
```

```
growth_leaders = df[df['March_Sales'] > 500].sort_values('Growth', ascending=False).head(3)

print("FASTEST GROWING PRODUCTS:")
print("=" * 40)
for index, row in growth_leaders.iterrows():
    print(f"{row['Product']}: {row['Growth']:.1f}% growth")
print("\n")
```

FASTEST GROWING PRODUCTS:

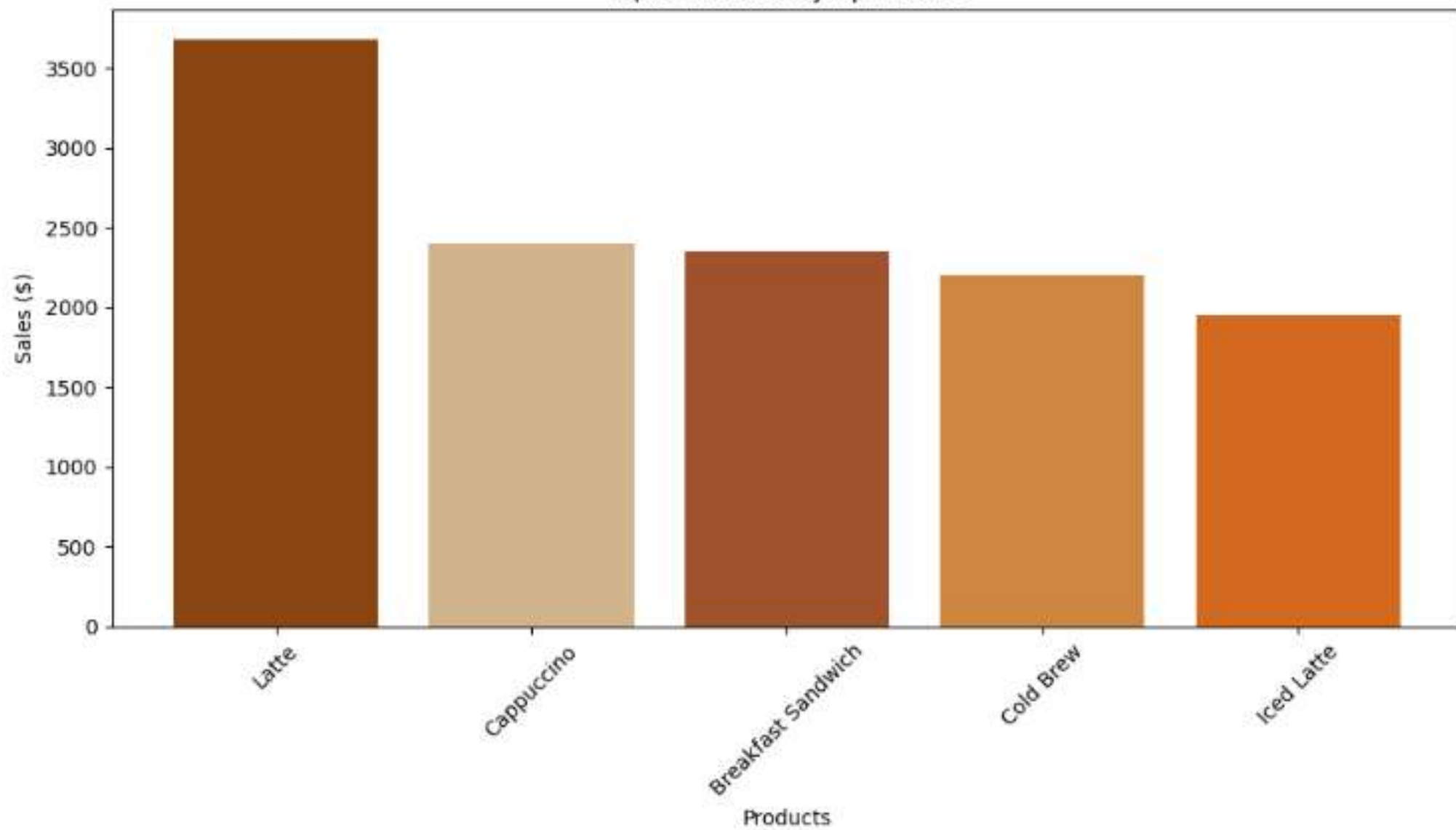
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Veggie Wrap: 20.8% growth

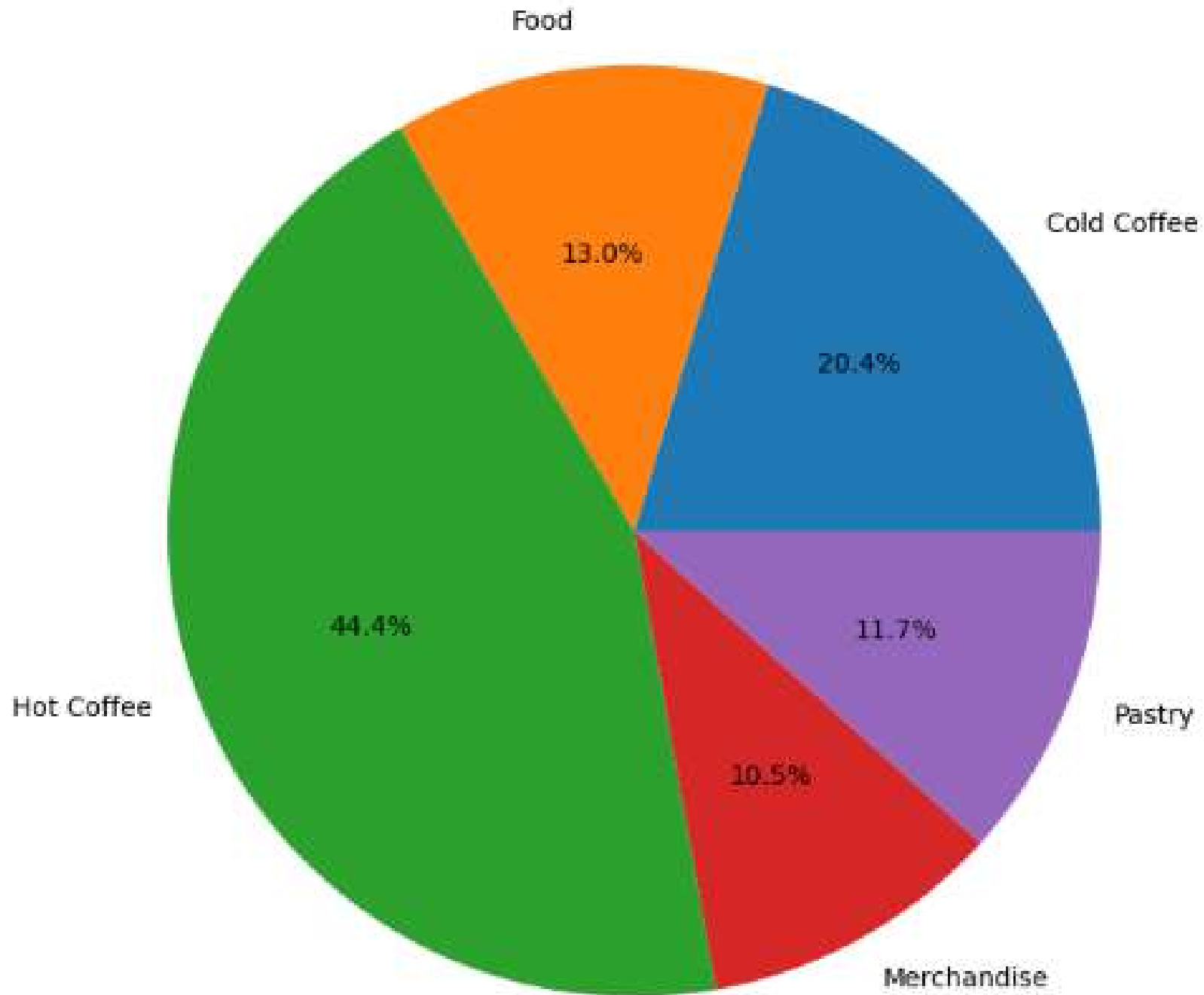
Scone: 20.0% growth

Coffee Beans: 20.0% growth

Top 5 Products by April Sales



Sales by Category



```
print("KEY INSIGHTS:")
print("=" * 40)
print(f"1. Our total sales grew by {total_growth:.1f}% from March to April")
print(f"2. The top selling product was {top_products.iloc[0]['Product']}")
print(f"3. The fastest growing product was {growth_leaders.iloc[0]['Product']}")
print(f"4. The {category_sales.idxmax()} category brings in the most revenue")
```

KEY INSIGHTS:

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1. Our total sales grew by 16.6% from March to April
2. The top selling product was Latte
3. The fastest growing product was Veggie Wrap
4. The Hot Coffee category brings in the most revenue